Your First 90 Days
with type 1 diabetes

A structured learning guide for newly diagnosed families
6 weeks · 18 topics · UK edition

Living Brilliantly with T1D

Educational only — not medical advice. Always follow your child's diabetes team.

Contents

Week 1 — Understanding Insulin

› What insulin actually does
› The two types of insulin your child uses
› Why insulin doses change

Week 2 — Carb Counting Basics

› Why carbs are the focus (not all food)
› How to count carbs practically
› The insulin-to-carb ratio explained

Week 3 — Exercise & Glucose

› Why exercise affects glucose so dramatically
› Practical strategies for sport days
› Getting active builds confidence — not just fitness

Week 4 — Overnight Glucose Patterns

› Why overnight glucose is uniquely challenging
› How CGM technology transforms overnight management
› When overnight glucose needs a plan adjustment

Week 5 — School & Social Life

› What schools are required to provide (England)
› Talking to your child about telling friends
› Building age-appropriate independence

Week 6 — Recognising & Managing Emergencies

› Recognising and treating hypoglycaemia
› When to use a glucagon kit
› Recognising DKA — a medical emergency

Everyone's child and care plan is different. Where this guide gives examples, your own diabetes team's advice always comes first.

WEEK 1

Understanding Insulin

Your most important foundation

What insulin actually does

Insulin is a hormone that acts like a key — unlocking cells so glucose can enter and be used for energy. Without it, glucose builds up in the bloodstream (hyperglycaemia) and cells go without fuel.

In Type 1, the immune system has destroyed the cells that produce insulin. This means your child's body cannot make any insulin at all — which is why insulin therapy is not optional; it's life-sustaining.

This is completely different from Type 2 diabetes, where the body still produces some insulin but uses it poorly.

The two types of insulin your child uses

Most children with type 1 diabetes use two main types of insulin:

Rapid-acting insulin (e.g., Novorapid, Humalog, Fiasp)

Given at mealtimes to 'cover' the glucose from food. Starts working in 10–15 minutes, peaks at 60–90 minutes, lasts 3–4 hours.

Long-acting (background) insulin (e.g., Lantus, Tresiba)

Given once or twice daily to provide a steady background level. Works slowly over 12–24 hours, keeping glucose stable between meals and overnight.

Think of it this way: background (basal) insulin is the baseline, and rapid-acting insulin is the response to food.

Why insulin doses change

You will notice that the same meal can produce different glucose outcomes on different days. This is normal — and expected.

Insulin needs change based on:

- Growth and puberty (increasing insulin resistance)
- Physical activity (increases insulin sensitivity)
- Illness (increases insulin resistance)
- Stress (increases insulin resistance)
- Sleep patterns
- Hormonal cycles in older children

This is why type 1 diabetes management is not a fixed formula — it is an ongoing adjustment. Your diabetes care team will help you review and update doses regularly.

WEEK 2

Carb Counting Basics

The skill that transforms meal management

Why carbs are the focus (not all food)

Carbohydrates are the main nutrient that raises blood glucose. Protein and fat have much smaller and more gradual effects.

This is why carb counting is the foundation of meal management in type 1 diabetes. When you know how many grams of carbohydrate are in a meal, you can calculate how much rapid-acting insulin is needed to keep glucose in range.

Foods that are mainly carbohydrate:

Bread, rice, pasta, potato, cereal, fruit, milk, yoghurt, sugar, juice, most snack foods.

Foods with very few carbs:

Meat, fish, chicken, eggs, cheese, most vegetables (except starchy ones), nuts, oils.

How to count carbs practically

You don't need to be exact — you need to be reasonably accurate.

Using food labels: On UK labels, look for the 'Carbohydrate' line — not just the 'of which sugars' line underneath it. Check whether the figure is per 100g or per portion, and weigh portions until you develop an eye for it.

Using apps: Apps like 'Carbs & Cals' (widely used in the UK) and 'MyFitnessPal' have extensive food databases with carb counts.

Common reference points to memorise:

- 1 slice of bread ≈ 15g carbs
- 1/2 cup cooked rice ≈ 20–25g carbs
- 1 medium apple ≈ 15–20g carbs
- 1 cup of milk ≈ 12g carbs
- 1 tablespoon of jam ≈ 10g carbs

Over time, estimating becomes intuitive. New families often say that within 6–8 weeks, they can estimate common meals without looking anything up.

The insulin-to-carb ratio explained

Your diabetes care team will give your child a personalised **insulin-to-carb ratio (ICR)** — for example, 1 unit of insulin for every 15g of carbohydrate.

This ratio tells you how much rapid-acting insulin to give for each meal.

Example:

ICR = 1:15

Meal contains 60g carbs

Dose = 60 ÷ 15 = 4 units

Different times of day may have different ratios — many children need more insulin per gram of carb at breakfast due to morning hormones.

Important: Do not change insulin-to-carb ratios without consulting your diabetes care team. If the ratio seems wrong, discuss it at your next appointment or call your team.

WEEK 3

Exercise & Glucose

Sport is wonderful — and manageable

Why exercise affects glucose so dramatically

Exercise changes glucose in ways that surprise many families — and understanding the mechanism makes it much less frightening.

During exercise (running, swimming, cycling, football):

Muscles can absorb glucose directly without insulin during activity. Glucose typically falls during and after exercise.

During anaerobic/intense exercise (sprinting, weightlifting, competitive racing):

The body releases stress hormones and stress hormones, which cause the liver to release stored glucose. This can temporarily RAISE glucose during very intense short bursts.

After exercise (delayed effect):

The body continues to store glucose in the muscles for up to 12 hours. This ongoing glucose absorption creates an extended hypoglycaemia risk — often most pronounced overnight.

Practical strategies for sport days

There is no single correct approach — every child, every sport, and every intensity level is different. Here are the most common strategies, to discuss with your diabetes care team:

Before exercise:

- Check glucose — target above 7–8 mmol/L before starting
- Have fast-acting carbs available
- Inform the coach

Reducing hypoglycaemia risk during sport:

- Reduce the insulin dose for the meal before sport (if planned in advance)
- Have a small carb snack before and/or during activity
- For pump users — discuss a temporary reduced background insulin rate with your team

After exercise:

- Check glucose before bed on sport days
- Consider a protein/carb bedtime snack
- Set CGM alerts at a higher threshold overnight

Important: These are general strategies. Your diabetes care team will help you develop a personalised sport protocol.

Getting active builds confidence — not just fitness

One of the most important things to communicate to your child is this: type 1 diabetes does not mean they can't do sport.

Some of the world's greatest athletes have type 1 diabetes:

- Jay Cutler — NFL quarterback
- Alexander Zverev — top-10 professional tennis player
- Numerous Olympic athletes

Regular physical activity is not just allowed with type 1 diabetes — it is actively encouraged. Exercise improves insulin sensitivity, cardiovascular health, and mental wellbeing.

The goal is not to avoid exercise because of glucose management. The goal is to learn to manage glucose so that exercise is always possible.

WEEK 4

Overnight Glucose Patterns

When to watch, when to act, when to sleep

Why overnight glucose is uniquely challenging

The overnight period presents unique challenges in type 1 diabetes management — and is often the source of the most parental anxiety.

Several factors affect glucose during sleep:

Growth hormone — Released in pulses during deep sleep, especially in children and adolescents. Causes temporary insulin resistance and glucose rises, typically between 2–8 AM. This is often called the 'dawn phenomenon'.

Delayed exercise effect — Activity during the day continues to lower glucose overnight as muscles replenish glycogen.

Long-acting insulin activity — Depending on when background insulin is given, its activity may peak or wane during sleep.

Fasting state — The liver releases small amounts of stored glucose during the overnight fast to maintain brain function.

How CGM technology transforms overnight management

Before continuous glucose monitors (CGMs), overnight management required parents to set alarms and do manual glucose checks — sometimes every 2 hours.

CGMs have transformed this completely.

What CGMs do overnight:

- Check glucose every 5 minutes, 24/7
- Alert your phone if glucose falls below your set threshold
- Alert if glucose is rising rapidly (early warning of a high)
- Allow you to see the overnight graph in the morning and identify patterns

For parents:

Many families set their low alert at a slightly higher threshold overnight (e.g., 5.5 mmol/L instead of 4.0) to give earlier warning.

Remote monitoring:

Apps like Dexcom Follow and LibreLinkUp allow parents to view their child's glucose remotely — including from a different bedroom.

When overnight glucose needs a plan adjustment

If you notice a consistent overnight pattern lasting more than 2–3 nights, bring it to your diabetes care team. Common adjustable patterns:

Consistent overnight highs:

May indicate the long-acting basal insulin dose needs adjustment, or growth hormone effect needs management.

Consistent overnight lows:

May indicate the long-acting dose is too high, or exercise from the previous day is having a sustained effect.

The 'roller coaster' overnight:

Sometimes glucose drops, triggers an alarm, gets treated with fast sugar, then rebounds high. This cycle is frustrating — discuss with your team a strategy to prevent it before it starts.

Important: Do not adjust basal insulin doses on your own without guidance. A small change to long-acting insulin has significant overnight effects.

School & Social Life

Building your child's independence

What schools are required to provide (England)

In England, schools have a legal duty to support pupils with medical conditions, including type 1 diabetes (Children and Families Act 2014), and must have regard to the Department for Education's statutory guidance. Your child should be able to:

- Check blood glucose or CGM at any time during the school day, including during class
- Treat a hypo immediately, without waiting for permission
- Access snacks and food when needed for medical reasons
- Have trained staff available for emergencies
- Keep their diabetes supplies accessible (not locked away)

What you should do:

Work with the school and your diabetes team to agree an **Individual Healthcare Plan (IHP)** — a written plan setting out what support your child needs, when, and from whom, including emergency procedures, trips and PE. Your child's diabetes team or school nurse can help with the plan and with training staff. The plan should be reviewed at least once a year, or sooner if your child's needs change.

Scotland, Wales and Northern Ireland have their own guidance — ask your diabetes team or Diabetes UK what applies where you live.

Talking to your child about telling friends

Whether and how to tell friends about type 1 diabetes is a deeply personal decision — and your child's preference should be respected.

Younger children (5–9):

Simple, matter-of-fact explanations work best. 'My body doesn't make insulin, so I need to give it to myself.' Most young children accept this without difficulty.

Older children and teens:

Some want their diagnosis known; others prefer privacy. Both are valid. What matters is that a trusted adult at school and a trusted friend knows the basics — in case of an emergency.

Script for your child:

'I have Type 1 Diabetes. It means my body can't make a hormone called insulin, so I take it myself. It's not contagious and I can do everything you can do.'

Children who feel open about type 1 diabetes typically report better social outcomes and less anxiety than those who feel they must hide it.

Building age-appropriate independence

The goal of type 1 diabetes management is not to remain dependent on parents forever. Gradually building your child's independence in their own care is one of the most important — and health-improving — things you can do.

Research shows:

Children who have age-appropriate ownership of their diabetes management have better long-term outcomes than those for whom parents manage everything.

Suggested milestones:

Ages 5–8: Recognise hypo/hyper symptoms and tell an adult. Know where their kit is.

Ages 8–12: Understand carb counting basics. Participate in checking glucose. Know how to treat a low independently.

Ages 12–15: Growing independence in dosing, with parental oversight. Check own glucose. Know sick day rules.

Ages 15+: Near-full management with regular family check-ins. Understand when to seek help.

This progression looks different for every child — base it on your child's maturity and confidence, not just age.

WEEK 6

Recognising & Managing Emergencies

Confidence in the moments that matter most

Recognising and treating hypoglycaemia

Low blood glucose (hypoglycaemia — 3.9 mmol/L or below) is the most common emergency in type 1 diabetes — and with the right knowledge, it is very manageable.

Common symptoms:

Shakiness, sweating, paleness, hunger, irritability, difficulty concentrating, glazed expression, confusion.

Important: Symptoms vary between children and can change over time. Some children develop 'hypoglycaemia unawareness' — they don't feel the symptoms until glucose is very low. CGMs significantly improve safety for these children.

Treating a hypo:

1. Give fast-acting carbohydrate — the amount depends on age: about 5g under 5, about 10g for ages 5–10, about 15g for ages 10–17, and never more than 20g (for example, for 10g: 3–4 glucose tablets, 120ml Lucozade, 2–3 jelly babies or 1 tube of GlucoGel)
2. Wait 10–15 minutes
3. Recheck glucose
4. If still below 5.6 mmol/L (or the level in your team's plan), repeat
5. If your team's plan says so — for example around exercise or at bedtime — follow with a small starchy snack

Chocolate, milk and biscuits act too slowly to treat a hypo. Your child's diabetes team may give different amounts — their plan comes first.

Never give food or drink to a child who is unconscious or unable to swallow safely.

When to use a glucagon kit

Glucagon is a hormone that rapidly raises blood glucose. Many children's diabetes teams prescribe a glucagon kit — ask your team whether your child should have one, and make sure everyone who cares for your child knows where it is and how to use it.

Use glucagon when:

- The child is unconscious or having a seizure due to low glucose
- The child cannot swallow safely
- Fast-acting glucose by mouth has not worked

Glucagon options:

- **Nasal glucagon (e.g. Baqsimi)** — Nasal powder. No injection required. Insert into one nostril and press.
- **Glucagon injection (e.g. GlucaGen HypoKit)** — Traditional injection kit. Requires mixing before use.

Your team will tell you which is right for your child and the dose to use.

After glucagon:

Call 999 if you have not already — glucagon is an emergency treatment, and your child should always be checked by a doctor afterwards. Tell your diabetes team.

Ensure:

Glucagon is available at school, with carers, and at home. Check expiry dates regularly. Ensure teachers and carers are trained to use it.

Recognising DKA — a medical emergency

Diabetic Ketoacidosis (DKA) is a serious medical emergency that occurs when the body has no insulin and begins breaking down fat for energy, producing acidic ketones.

DKA warning signs:

- Persistent high glucose (above 14 mmol/L) that is not coming down
- Vomiting or tummy pain
- Deep, rapid breathing
- Fruity or acetone-like breath
- Extreme tiredness or confusion
- Blood ketones of 1.5 mmol/L or above (3.0 or above means DKA is likely)

Action:

Go to your nearest A&E now, or call 999 if your child is drowsy, confused or struggling to breathe. Do not wait.

At blood ketones of 1.5 mmol/L or above, speak to your diabetes team straight away.

Prevention:

- Never skip insulin — even if the child is unwell and not eating
- Check ketones whenever your child is unwell, and any time glucose is above 14 mmol/L for more than 2 hours
- Have a sick day plan from your diabetes team
- Check pump infusion sites regularly if using a pump

Important: DKA can develop in hours. When in doubt, seek help early.

Emergency numbers and help (UK)

Emergency (ambulance, fire, police)	**999 or 112**
NHS 111 — urgent advice (England, Scotland, Wales)	**111**
Your diabetes team — 24-hour line	_____
Your local A&E	_____
Diabetes UK helpline (Mon–Fri, 9am–6pm)	**0345 123 2399**
Samaritans (free, 24/7)	**116 123**
Childline (under 19, free, 24/7)	**0800 1111**
Shout (free text, 24/7)	**Text SHOUT to 85258**

In Northern Ireland, NHS 111 is not available in the same way — use your hospital's out-of-hours number or GP out-of-hours service.

Medical disclaimer

This guide is for educational and informational purposes only. It does not constitute medical advice, diagnosis, or treatment. Always seek guidance from your qualified healthcare provider — including your paediatric diabetes team, diabetes specialist nurse, and GP — for any questions about your child's medical condition, treatment plan, or medications.

Never disregard professional medical advice or delay seeking it because of something you have read in this guide. Insulin doses, ratios and protocols mentioned here are illustrative examples only — your child's actual doses must be set by your diabetes team.

In an emergency, call 999.